
Rethinking Importance Sampling in LLM Policy Optimization: A Cumulative Token Perspective

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Abstract

Reinforcement learning, including reinforcement learning with verifiable rewards (RLVR), has emerged as a powerful approach for LLM post-training. Central to these approaches is the design of the importance sampling (IS) ratio used in off-policy policy-gradient estimation. Existing methods face a fundamental bias-variance dilemma: token-level IS ratios, as adopted by PPO [Schulman et al., 2017] and GRPO [Shao et al., 2024], introduce bias by ignoring prefix state distribution mismatch; full sequence ratios provide exact trajectory-level correction but suffer from high variance due to the multiplicative accumulation of per-token ratios, while GSPO [Zheng et al., 2025] improves numerical stability via length normalization at the cost of deviating from the exact full-sequence IS correction. In this work, we identify the cumulative token IS ratio, the product of per-token ratios up to position t , as a theoretically principled solution to this dilemma. We prove that, under the token-level policy-gradient formulation, this ratio provides an unbiased prefix correction for each token-level gradient term and has strictly lower variance than the full sequence ratio. Building on this insight, we propose CTPO (Cumulative Token Policy Optimization), which combines the cumulative token IS ratio with position-adaptive clipping that scales log-space clip bounds according to the natural $\sqrt{t}$ growth of the cumulative log-ratio. This yields more consistent regularization across token positions. We implement and evaluate CTPO in the tool-integrated reasoning setting on several challenging mathematical reasoning benchmarks, achieving the best average performance across both model scales compared with strong GRPO and GSPO baselines. Code will be available at <https://github.com/horizon-llm/CTPO>.

1 Introduction

Reinforcement learning has emerged as a powerful paradigm for LLM post-training, achieving remarkable success on tasks with verifiable rewards such as mathematical reasoning and code generation. Landmark systems such as OpenAI o1 [Jaech et al., 2024] and DeepSeek-R1 [Guo et al., 2025] have demonstrated that reinforcement learning can dramatically enhance the reasoning capabilities of large language models. A standard algorithmic framework underlying these advances is policy gradient optimization, with Group Relative Policy Optimization (GRPO) [Shao et al., 2024] serving as a representative example. Subsequent works have proposed numerous variants of GRPO, including DAPO [Yu et al., 2025] and Dr. GRPO [Liu et al., 2025], addressing issues such as sampling efficiency and reward normalization bias.

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A central challenge in these policy gradient methods is handling the off-policy nature of training, where trajectories are collected from a behavior policy π_b but used to update a different target policy π_θ . Importance sampling (IS) is the standard tool for correcting this distribution mismatch. However, existing methods adopt IS ratio formulations with fundamental bias-variance limitations. GRPO uses token-level IS ratios $r_t = \pi_\theta(a_t | s_t) / \pi_b(a_t | s_t)$, which account only for the current-token action probability while ignoring the preceding state distribution mismatch, introducing a systematic bias into the gradient estimate. A full sequence ratio would correct the trajectory-level mismatch, but it suffers from high variance due to the multiplicative accumulation of per-token ratios across the entire response. GSPO [Zheng et al., 2025] improves numerical stability by using a length-normalized sequence-level ratio, equivalently the geometric mean of per-token ratios, but this normalized ratio no longer corresponds to the exact full-sequence IS correction and is therefore biased as an IS correction.

In this work, we identify the cumulative token IS ratio as a theoretically principled solution to this bias-variance dilemma. The cumulative token IS ratio at position t is defined as the product of per-token ratios up to position t , i.e., $\rho_t^{\text{cum}} = \prod_{t'=1}^t r_{t'}$. We prove that, under the token-level policy-gradient formulation, this ratio provides an unbiased prefix correction for each token-level gradient term, since it correctly accounts for the prefix trajectory likelihood under both policies. At the same time, it has strictly lower variance than the full sequence ratio, as suffix tokens beyond position t have unit expectation under the behavior policy and contribute only unnecessary variance without reducing bias. This makes the cumulative token IS ratio a principled middle ground: it preserves the exact prefix correction needed at each token position while avoiding the unnecessary suffix variance of the full sequence ratio.

Building on this insight, we further identify a practical challenge that arises when applying clipping to the cumulative token IS ratio. Since $\log \rho_t^{\text{cum}} = \sum_{t'=1}^t \log r_{t'}$ accumulates per-token log-ratios along the sequence, its variance grows linearly with position: $\text{Var}(\log \rho_t^{\text{cum}}) = t\sigma^2$. As a result, a fixed clipping range leads to inconsistent regularization across token positions — early tokens are rarely clipped while late tokens are clipped far more frequently. We empirically observe this log-space variance growth during training and propose a position-adaptive clipping strategy that scales the clipping thresholds proportionally to $\sqrt{t}$, matching the natural standard deviation growth of the cumulative log-ratio and providing more consistent regularization across token positions.

We instantiate these ideas in CTPO (Cumulative Token Policy Optimization) and evaluate it in the tool-integrated reasoning (TIR) setting [Xue et al., 2025], where the model iteratively generates and executes Python code in a sandboxed environment to solve challenging mathematical problems. Our contributions are summarized as follows:

- **Cumulative Token IS Ratio.** We identify the cumulative token IS ratio as a theoretically principled solution to the bias-variance dilemma in off-policy LLM post-training. Under the token-level policy-gradient formulation, we prove that it provides an unbiased prefix correction for each gradient term and has strictly lower variance than the full sequence ratio.
- **Position-Adaptive Clipping.** We propose a position-adaptive clipping strategy based on the observation that $\text{Var}(\log \rho_t^{\text{cum}})$ grows linearly with token position. By scaling the clipping thresholds proportionally to $\sqrt{t}$, our strategy provides more uniform regularization across token positions compared to fixed clipping.
- **Empirical Validation.** We evaluate CTPO in the tool-integrated reasoning setting on challenging mathematical benchmarks, where it achieves the best average performance across both model scales compared with strong GRPO and GSPO baselines. Ablation studies further confirm the contribution of position-adaptive clipping to overall performance.

2 Preliminaries

We first review the token-level MDP formulation underlying LLM generation, then present GRPO and GSPO as two representative baselines that adopt different importance sampling strategies.

Token-level MDP. LLM text generation can be formulated as a finite-horizon Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{V}, P, R, H)$, where the state s_t at step t is the concatenation of the prompt x and all previously generated tokens $(a_1, \dots, a_{t-1})$, the action $a_t \in \mathcal{V}$ is the next token drawn from the

policy $\pi_\theta(\cdot \mid s_t)$, and the transition is deterministic: $s_{t+1} = s_t \circ a_t$. A scalar reward $R(\tau)$ is assigned to the complete trajectory $\tau = (s_1, a_1, \dots, s_H, a_H)$ upon termination.

Policy Gradient and Importance Sampling. The learning objective is to maximize the expected reward $J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)]$. By the policy gradient theorem, the gradient is:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=1}^H A_t(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t \mid s_t) \right],$$

where $A_t(s_t, a_t)$ is the advantage function at position t . In practice, trajectories are collected from a fixed behavior policy $\pi_b = \pi_{\theta_{\text{old}}}$ and reused across multiple gradient steps. To correct for this distribution mismatch, importance sampling (IS) yields:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_b} \left[\sum_{t=1}^H \rho_t A_t(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t \mid s_t) \right],$$

where ρ_t is the IS ratio at position t . The choice of ρ_t is central to the bias-variance tradeoff and is the main distinction between existing algorithms.

2.1 Group Relative Policy Optimization (GRPO)

GRPO [Shao et al., 2024] eliminates the value network of PPO by estimating advantages through group-relative reward normalization. For each prompt x , the behavior policy $\pi_{\theta_{\text{old}}}$ generates G responses $\{o_i\}_{i=1}^G$. The advantage of response o_i is computed as:

$$A_i = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^G)}{\text{std}(\{R_j\}_{j=1}^G)},$$

where $R_i = \mathcal{R}(x, o_i)$ is the scalar reward assigned to the entire response o_i . Since the reward is only available at the sequence level, GRPO assigns the same advantage to every token in o_i , i.e., $A_t(s_t, a_t) = A_i$ for all $t \in \{1, \dots, |o_i|\}$. GRPO then maximizes the following clipped surrogate objective:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min(r_{i,t} A_i, \text{clip}(r_{i,t}, 1 - \varepsilon, 1 + \varepsilon) A_i) \right],$$

where the token-level IS ratio is:

$$r_{i,t} = \frac{\pi_\theta(o_{i,t} \mid x, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid x, o_{i,<t})}.$$

Bias of the token-level IS ratio. While $r_{i,t}$ is straightforward to compute, it constitutes a *biased* estimator of the true IS weight at position t . The theoretically correct IS ratio for correcting the distribution mismatch of the state-action pair (s_t, a_t) must account for both the probability of reaching s_t and sampling a_t under π_θ versus π_b , which involves the prefix trajectory up to position t :

$$\rho_t^* = \frac{\pi_\theta(o_{i,1:t} \mid x)}{\pi_b(o_{i,1:t} \mid x)} = \prod_{t'=1}^t r_{i,t'}.$$

GRPO uses only the current-token ratio $r_{i,t}$, discarding the prefix $\prod_{t'=1}^{t-1} r_{i,t'}$. This simplification introduces a systematic bias in the gradient estimate, though it yields lower variance.

2.2 Group Sequence Policy Optimization (GSPO)

A natural way to address the token-level bias is to define the IS ratio at the sequence level. The full sequence ratio for response o_i is:

$$\rho_i^{\text{seq}} = \frac{\pi_\theta(o_i \mid x)}{\pi_b(o_i \mid x)} = \prod_{t=1}^{|o_i|} r_{i,t}.$$

This full sequence ratio corrects the trajectory-level distribution mismatch, but it can have high variance due to the multiplicative accumulation of per-token ratios across the entire response. To mitigate the high variance, GSPO [Zheng et al., 2025] uses a length-normalized sequence-level ratio:

$$\rho_i^{\text{GSPO}} = \left(\frac{\pi_\theta(o_i | x)}{\pi_b(o_i | x)} \right)^{1/|o_i|} = \left(\prod_{t=1}^{|o_i|} r_{i,t} \right)^{1/|o_i|}.$$

This ratio is equivalently the geometric mean of the per-token ratios within the sequence. It is applied uniformly to the sequence, and GSPO clips at the sequence level rather than the token level:

$$\mathcal{J}_{\text{GSPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \min(\rho_i^{\text{GSPO}} A_i, \text{clip}(\rho_i^{\text{GSPO}}, 1 - \varepsilon, 1 + \varepsilon) A_i) \right].$$

Length normalization reduces the scale of the IS ratio and improves numerical stability, but the resulting ratio no longer corresponds to the exact full-sequence IS correction.

In summary, the token-level ratio and the full sequence ratio represent two ends of the bias-variance spectrum: the token-level ratio is biased but low-variance, while the full sequence ratio is unbiased but high-variance. GSPO reduces variance through length normalization, but its normalized sequence-level ratio is still biased as an IS correction. In the next section, we show that the cumulative token IS ratio resolves this dilemma by achieving both unbiasedness and lower variance than the full sequence formulation.

3 Cumulative Token Policy Optimization

3.1 Cumulative Token Importance Ratio

Recall that the policy gradient requires an expectation under the current policy π_θ , but in practice trajectories are sampled from the behavior policy π_b . Throughout this section, we analyze importance-ratio correction for a generic token-level advantage function $A_t(s_t, a_t)$. To correct the distribution mismatch for the state-action pair (s_t, a_t) at position t , the IS weight must account for the probability of reaching s_t and sampling a_t under π_θ relative to π_b . Since the transition dynamics are deterministic, this reduces to the ratio of the prefix trajectory likelihoods:

$$\rho_t^{\text{cum}} = \frac{\pi_\theta(a_{1:t} | x)}{\pi_b(a_{1:t} | x)} = \prod_{t'=1}^t \frac{\pi_\theta(a_{t'} | s_{t'})}{\pi_b(a_{t'} | s_{t'})} = \prod_{t'=1}^t r_{t'}.$$

We term this the *cumulative token importance ratio*: unlike GRPO, which uses only the current-token ratio r_t and ignores the prefix trajectory, and unlike the full sequence ratio $\prod_{t'=1}^H r_{t'}$, which includes the entire response, ρ_t^{cum} accumulates per-token ratios only up to position t , naturally matching the temporal structure of the MDP. Intuitively, ρ_t^{cum} asks: how much more or less likely is it that π_θ would have produced the prefix $(a_1, \dots, a_t)$ compared to π_b ? This is precisely the prefix correction factor needed for the token-level policy-gradient term at position t , as we formalize in the following proposition.

Proposition 1 (Unbiasedness of Cumulative Token IS Ratio). *Under the token-level MDP formulation, let $A_t(s_t, a_t)$ denote the token-level advantage function, the gradient estimator using the cumulative token IS ratio is unbiased:*

$$\mathbb{E}_{\tau \sim \pi_b} \left[\sum_{t=1}^H \rho_t^{\text{cum}} A_t(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t) \right] = \nabla_\theta J(\theta).$$

The proof proceeds by applying a change of measure to the prefix trajectory $a_{1:t}$, after which marginalizing over the suffix $a_{t+1:H}$ introduces no additional correction factor since both A_t and $\nabla_\theta \log \pi_\theta(a_t | s_t)$ depend only on $(x, a_{1:t})$. The full proof is provided in Appendix A.1.

3.2 Bias-Variance Analysis

We now analyze the bias and variance properties of the considered IS ratio formulations. As we will show, the token-level ratio r_t sacrifices prefix correction for low variance, the full sequence ratio ρ^{seq} is unbiased but suffers from high variance, and the cumulative token ratio ρ_t^{cum} preserves the exact prefix correction while avoiding unnecessary suffix variance.

Bias of the token-level ratio. The token-level ratio r_t used by GRPO is a biased estimator of the true IS weight. The correct IS weight for position t must account for the probability of reaching state s_t under π_θ versus π_b , which requires the prefix ratio ρ_t^{cum} . Formally, for $t > 1$,

$$\mathbb{E}_{\tau \sim \pi_b} [r_t A_t(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t)] \neq \mathbb{E}_{\tau \sim \pi_\theta} [A_t(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t)]$$

in general, since r_t only corrects the current-token action probability and omits the prefix correction $\prod_{t'=1}^{t-1} r_{t'}$ for the state distribution mismatch.

Variance of the sequence-level ratio. The sequence-level ratio $\rho^{\text{seq}} = \prod_{t'=1}^H r_{t'}$ is unbiased but incurs unnecessarily high variance. To see this, observe that for any position t :

$$\rho^{\text{seq}} = \rho_t^{\text{cum}} \cdot \underbrace{\prod_{t'=t+1}^H r_{t'}}_{\epsilon_t},$$

where $\epsilon_t = \prod_{t'=t+1}^H r_{t'}$ is the suffix ratio from position $t+1$ to H . By the likelihood ratio identity:

$$\mathbb{E}_{\pi_b}[\epsilon_t | s_t, a_t] = 1,$$

so ϵ_t does not contribute to correcting the distribution mismatch at position t , yet it inflates the variance of the estimator. This motivates replacing ρ^{seq} with ρ_t^{cum} , which retains all information necessary for unbiasedness while discarding the variance-inflating suffix.

We formalize this variance reduction in the following proposition.

Proposition 2 (Variance Reduction of Cumulative Token IS Ratio). *The following statements hold regarding the variance of the IS ratios. (i) For any $t < H$, if π_θ and π_b differ on a π_b reachable future state, then:*

$$\text{Var}(\rho^{\text{seq}}) > \text{Var}(\rho_t^{\text{cum}}),$$

(ii) (Independence assumption) Further assume the per-token ratios $r_{t'}$ are independent across positions, and let $\chi_{t'}^2 = \mathbb{E}_{\pi_b} [\chi^2(\pi_\theta(\cdot | s_{t'}) || \pi_b(\cdot | s_{t'}))]$ denote the averaged local χ^2 divergence at position t' . Then:

$$\text{Var}(\rho_t^{\text{cum}}) = \prod_{t'=1}^t (1 + \chi_{t'}^2) - 1, \quad \text{Var}(\rho^{\text{seq}}) = \prod_{t'=1}^H (1 + \chi_{t'}^2) - 1,$$

and consequently:

$$\frac{\text{Var}(\rho^{\text{seq}})}{\text{Var}(\rho_t^{\text{cum}})} = \frac{\prod_{t'=1}^H (1 + \chi_{t'}^2) - 1}{\prod_{t'=1}^t (1 + \chi_{t'}^2) - 1}.$$

The proof is deferred to Appendix A.2. Part (i) holds without any distributional assumptions, showing that the cumulative token IS ratio has strictly lower variance than the full sequence ratio whenever the future token ratios after position t vary under the behavior policy. Part (ii) further quantifies this reduction. As a concrete example, if all positions share the same averaged local divergence $\chi_{t'}^2 = \delta$, the variance ratio equals $\frac{(1+\delta)^H - 1}{(1+\delta)^t - 1}$. In the near on-policy regime where π_θ and π_b are close, $\delta \rightarrow 0$ and the ratio converges to H/t , meaning tokens at early positions benefit the most from using ρ_t^{cum} over ρ^{seq} . As the degree of off-policy increases, δ grows and the ratio scales as $(1+\delta)^{H-t}$, increasing exponentially with the remaining sequence length $H-t$. This advantage grows with sequence length H , making the cumulative token IS ratio particularly well-suited for tasks that require extended generation.

We summarize the bias-variance properties of the considered IS ratio formulations in Table 1. GRPO uses a token-level ratio, which has low variance but omits prefix correction. The full sequence ratio is unbiased but suffers from high variance. GSPO reduces the scale of the importance weight through length normalization, but the normalized ratio is biased as an IS correction. CTPO achieves the desired combination among these formulations: it preserves the exact prefix correction while having strictly lower variance than the full sequence ratio.

Table 1: Comparison of importance sampling ratio formulations in terms of bias and variance.

Method	IS Ratio	Unbiased	Variance
GRPO	r_t	✗	Low
Full sequence ratio	$\prod_{t'=1}^H r_{t'}$	✓	High
GSPO (length-normalized)	$\left(\prod_{t'=1}^H r_{t'}\right)^{1/H}$	✗	Reduced scale
CTPO (ours)	$\prod_{t'=1}^t r_{t'}$	✓	Lower than full sequence

3.3 Position-Adaptive Clipping

Clipping is a standard technique in PPO and GRPO to stabilize training by constraining the IS ratio within a fixed range $[1 - \varepsilon, 1 + \varepsilon]$, preventing excessively large policy updates. However, directly applying this fixed clipping range to ρ_t^{cum} is inadequate. Recall that $\log \rho_t^{\text{cum}} = \sum_{t'=1}^t \log r_{t'}$ accumulates the per-token log-ratios along the generated sequence, and as these shifts accumulate, its magnitude tends to grow with position t . To see this more precisely, assuming the per-token log-ratios are independent with variance σ^2 , the variance of the cumulative sum grows linearly with position:

$$\text{Var}(\log \rho_t^{\text{cum}}) = t\sigma^2,$$

causing early tokens to be rarely clipped while later tokens with larger cumulative deviations are clipped far more frequently, leading to inconsistent regularization across positions.

To address this, we propose scaling the clipping thresholds in log-space proportionally to the standard deviation of $\log \rho_t^{\text{cum}}$. Specifically, we define position-adaptive clipping thresholds as:

$$\varepsilon_{\text{high}}(t) = \varepsilon_{\text{high}} \cdot t^p, \quad \varepsilon_{\text{low}}(t) = \varepsilon_{\text{low}} \cdot t^p,$$

where $\varepsilon_{\text{high}}, \varepsilon_{\text{low}} > 0$ are base thresholds and $p > 0$ is the scaling exponent, yielding the position-dependent trust region:

$$\rho_t^{\text{cum}} \in \left[e^{-\varepsilon_{\text{low}}(t)}, e^{\varepsilon_{\text{high}}(t)} \right].$$

With $p = 0.5$, the clipping thresholds grow as $\sqrt{t}$, matching the standard deviation growth of $\log \rho_t^{\text{cum}}$ under the independence assumption, providing more consistent regularization across positions.

In practice, following GRPO, we avoid training a separate critic network and use an outcome-level group-relative reward as a surrogate advantage estimate. Specifically, for each sampled response o_i , we compute a scalar advantage A_i from the normalized outcome reward and apply it uniformly across all token positions. Combining this practical advantage estimation strategy with the cumulative token IS ratio and position-adaptive clipping, the final CTPO objective is:

$$\mathcal{J}_{\text{CTPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(\rho_{i,t}^{\text{cum}} A_i, \text{clip} \left(\rho_{i,t}^{\text{cum}}, e^{-\varepsilon_{\text{low}}(t)}, e^{\varepsilon_{\text{high}}(t)} \right) A_i \right) \right].$$

4 Experiments

4.1 Experimental Setup

Task and Models. We focus on the tool-integrated reasoning (TIR) setting [Xue et al., 2025], where the model iteratively generates Python code, executes it in a sandboxed Python interpreter, and uses the returned output to inform subsequent reasoning steps. Each trajectory consists of up to 5 turns of model-environment interaction, with a maximum response length of 8,000 tokens per trajectory. We conduct experiments on two model scales: Qwen3-4B and Qwen3-14B [Yang et al., 2025].

Datasets and Benchmarks. We train on DeepScaleR [Luo et al., 2025], a curated dataset of approximately 40K mathematics problems. We evaluate on four challenging competition-level mathematical reasoning benchmarks: AIME 2025², AIME 2026³, HMMT 2025⁴, and BRUMO

²<https://huggingface.co/datasets/math-ai/aime25>

³<https://huggingface.co/datasets/math-ai/aime26>

⁴https://huggingface.co/datasets/MathArena/hmmt_nov_2025

Table 2: Main results on competition-level mathematical reasoning benchmarks in the tool-integrated reasoning setting. All models are evaluated with avg@32. CTPO achieves the best average performance across both model scales.

Base Model	Method	AIME 25	AIME 26	BRUMO 25	HMMT 25	Avg
Qwen3-4B	Base	1.9	3.4	6.3	4.1	3.9
	GRPO	43.0	44.1	53.1	32.6	43.2
	GSPO	49.2	47.6	56.8	37.0	47.7
	CTPO (Ours)	53.5	52.1	59.7	40.4	51.4
Qwen3-14B	Base	4.3	6.9	7.0	6.0	6.1
	GRPO	55.3	49.8	61.9	51.5	54.6
	GSPO	56.4	56.1	63.3	46.0	55.5
	CTPO (Ours)	65.0	59.1	63.0	48.0	58.8

2025⁵, spanning a range of olympiad-level competition problems. For each benchmark we report avg@32, where we sample 32 responses per problem and average the pass rate.

Baselines and Implementation. We compare CTPO against two baselines representing different IS ratio designs: GRPO [Shao et al., 2024], which adopts token-level IS ratios, and GSPO [Zheng et al., 2025], which adopts a length-normalized sequence-level IS ratio. To isolate the effect of IS ratio design on downstream performance, we keep all other training configurations identical across methods: a prompt batch size of 512, 8 responses per prompt, and a learning rate of 1×10^{-6} , implemented using the VERL framework [Sheng et al., 2025]. For CTPO, we set the position-adaptive clipping hyperparameters to $\varepsilon_{\text{low}} = 0.025$ and $\varepsilon_{\text{high}} = 0.05$. All experiments are conducted on a single node of 8 NVIDIA H100 or H200 GPUs.

4.2 Main Results

Table 2 presents the main results of CTPO against GRPO and GSPO on four competition-level mathematical reasoning benchmarks in the tool-integrated reasoning setting.

CTPO consistently achieves the best average performance across both model scales. On Qwen3-4B, CTPO outperforms the strongest baseline GSPO by 3.7 points on average (51.4 vs. 47.7), a relative improvement of 7.8%. On Qwen3-14B, CTPO outperforms GSPO by 3.3 points (58.8 vs. 55.5), a relative improvement of 5.9%. Gains are observed consistently across both model scales, demonstrating that the cumulative token IS ratio provides a reliable improvement over strong token-level and length-normalized sequence-level baselines.

These results support our central claim: by preserving the exact prefix correction needed at each token position while avoiding unnecessary suffix variance, the cumulative token IS ratio translates its theoretical advantage into consistent average improvements across challenging competition-level benchmarks.

4.3 Cumulative IS Ratio Variance and Clipping Behavior

In this subsection, we analyze the dynamics of the cumulative token IS ratio ρ_t^{cum} during training to empirically examine its position-dependent variance growth and motivate the position-adaptive clipping design. All analyses in this section are based on Qwen3-4B.

Variance growth of $\log \rho_t^{\text{cum}}$. The top row of Figure 1 shows the empirical standard deviation of $\log \rho_t^{\text{cum}}$ as a function of token position t across training steps 50, 100, and 150. In all cases, the empirical std closely follows the fitted curve $\hat{\sigma}\sqrt{t}$, consistent with the theoretical calculation $\text{Var}(\log \rho_t^{\text{cum}}) = t\sigma^2$ from Section 3.3. This variance growth with position confirms that a fixed clipping range becomes increasingly misaligned as token position grows.

⁵https://huggingface.co/datasets/MathArena/brumo_2025

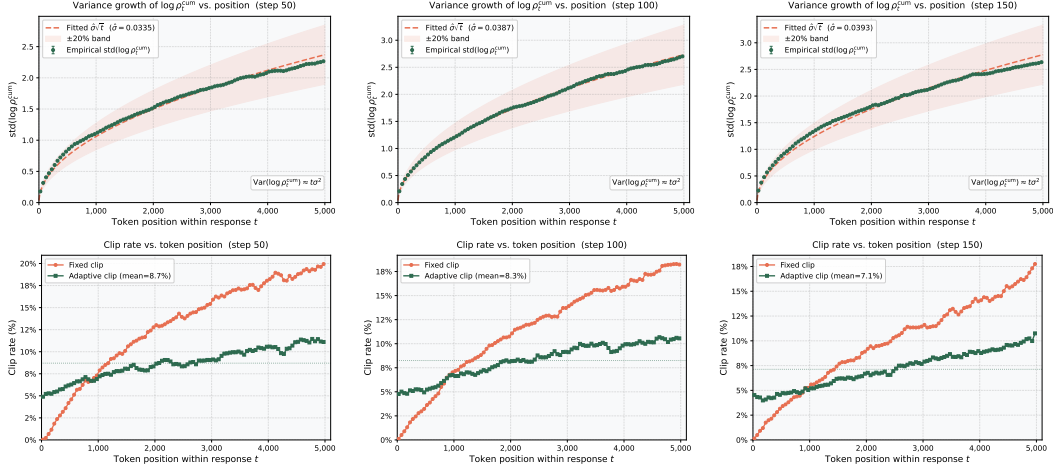


Figure 1: Analysis of $\log \rho_t^{\text{cum}}$ across training steps 50, 100, and 150. **Top row:** Empirical standard deviation of $\log \rho_t^{\text{cum}}$ vs. token position t , fitted with $\hat{\sigma}\sqrt{t}$, confirming the log-space variance growth discussed in Section 3.3. **Bottom row:** Clip rate vs. position under fixed clipping (ratio $\in [0.5, 5]$) and adaptive clipping. The fixed clip rate grows monotonically with t , while adaptive clipping maintains a substantially more uniform clip rate across positions.

Table 3: Ablation study on position-adaptive clipping vs. fixed clipping. Position-adaptive clipping consistently outperforms fixed clipping across all benchmarks.

Method	AIME 25	AIME 26	BRUMO 25	HMMT 25	Avg
CTPO w/ fixed clip	55.5	57.5	62.6	47.1	55.7
CTPO (Ours)	65.0	59.1	63.0	48.0	58.8

Clip rate under fixed vs. adaptive clipping. The bottom row of Figure 1 compares the per-position clip rate under fixed clipping (ratio $\in [0.5, 5]$) and adaptive clipping across the same training steps. The fixed clip rate increases monotonically with position t , reaching up to 20% for late tokens while remaining near 0% for early tokens. This positional imbalance disproportionately suppresses gradient updates from late tokens, an issue that is further amplified in extended generation tasks. In contrast, adaptive clipping maintains a substantially more uniform clip rate across all positions, consistently around 5–10% regardless of token position or training step. This demonstrates that position-adaptive clipping effectively compensates for the variance growth of $\log \rho_t^{\text{cum}}$.

4.4 Ablation Study

Table 3 ablates the contribution of position-adaptive clipping on Qwen3-14B by replacing it with a fixed clipping range while keeping all other components of CTPO unchanged. Position-adaptive clipping outperforms fixed clipping by 3.1 points on average (58.8 vs. 55.7), with consistent gains across all four benchmarks. These results indicate that the position-dependent variance growth of $\log \rho_t^{\text{cum}}$ has practical consequences for training, and that scaling the clipping thresholds proportionally to $\sqrt{t}$ provides a meaningful benefit beyond the theoretical motivation.

4.5 Training Dynamics

Figure 2 shows the training dynamics of GRPO, GSPO, and CTPO on Qwen3-14B. CTPO achieves higher AIME 2025 accuracy than both baselines throughout training, with the performance gap becoming more pronounced in the later stages of training. The clip fraction of CTPO lies between that of GRPO and GSPO and remains stable throughout training, indicating that position-adaptive clipping does not introduce training instability. All three methods also exhibit similar response length growth, suggesting that CTPO does not noticeably alter generation length dynamics.

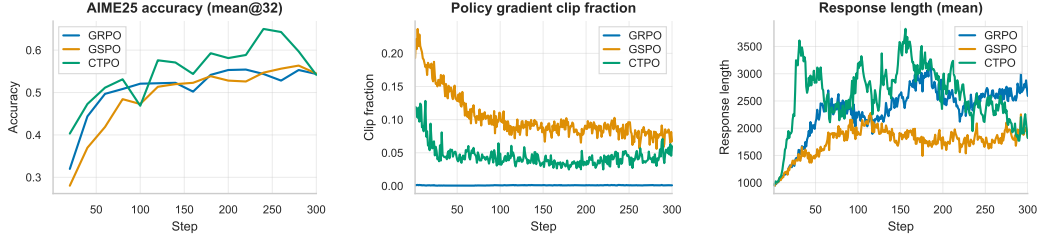


Figure 2: Training dynamics of GRPO, GSPO, and CTPO. **Left:** AIME 2025 accuracy (avg@32) throughout training. **Middle:** Policy gradient clip fraction, measured as the fraction of tokens whose method-specific IS ratio exceeds the corresponding clipping threshold. **Right:** Mean response length throughout training.

5 Related Work

Importance Sampling in Reinforcement Learning. Importance sampling (IS) has a long history in off-policy reinforcement learning as a tool for correcting distributional mismatch between the behavior and target policies. Early work by Precup et al. [2000] established foundational IS-based off-policy evaluation methods, introducing per-decision importance sampling that exploits the temporal structure of MDPs to reduce variance by discarding future terms irrelevant to the current decision. Variance reduction techniques such as control variates [Greensmith et al., 2004] and weighted importance sampling [Thomas, 2015] have also been widely studied in this context. For off-policy value evaluation, doubly robust estimators [Jiang and Li, 2016] combine importance sampling with value function estimates to reduce variance while preserving unbiasedness under standard conditions. Our work builds on these classical ideas and applies them to the specific structure of LLM token generation, where the token-level MDP formulation naturally motivates the cumulative token IS ratio as a theoretically principled per-decision correction.

Reinforcement Learning for LLM Post-Training. Reinforcement learning has become a cornerstone of LLM post-training since the advent of RLHF [Ouyang et al., 2022]. PPO [Schulman et al., 2017] was among the first algorithms applied in this setting to optimize a KL-regularized reward objective [Bai et al., 2022]. Rafailov et al. [2024] introduced DPO as a simpler alternative that directly optimizes a Bradley-Terry-derived loss without explicit reward modeling, spawning a large family of variants [Ethayarajh et al., 2024, Meng et al., 2024, Dong et al., 2024]. A parallel line of work relaxes the Bradley-Terry assumption entirely [Azar et al., 2024, Munos et al., 2023, Ye et al., 2024, Wu et al., 2024, Zhang et al., 2024, 2025b]. More recently, reinforcement learning has achieved notable success in enhancing the reasoning capabilities of LLMs [Jaech et al., 2024, Guo et al., 2025]. A representative example is GRPO [Shao et al., 2024], which adopts token-level IS ratios and estimates advantages through group-relative reward normalization. A number of subsequent works build on the GRPO framework while retaining token-level IS ratios, improving different components of the training pipeline. DAPO [Yu et al., 2025] introduces dynamic sampling, Dr. GRPO [Liu et al., 2025] corrects length normalization bias, SAPO [Gao et al., 2025] replaces hard clipping with a soft gating mechanism, and AR3PO [Zhang et al., 2025a] improves sampling efficiency via adaptive rollout and response reuse. In contrast, GSPO [Zheng et al., 2025] adopts a length-normalized sequence-level ratio, equivalently the geometric mean of per-token ratios, to improve numerical stability, but this normalized ratio no longer corresponds to the exact full-sequence IS correction. Our work addresses this gap by proposing the cumulative token IS ratio, which preserves the exact prefix correction needed for token-level policy-gradient terms while avoiding unnecessary suffix variance from the full sequence ratio.

6 Conclusion

We introduced CTPO, which combines the cumulative token IS ratio with position-adaptive clipping to provide principled prefix correction and more consistent regularization, yielding empirical improvements in tool-integrated mathematical reasoning. In the future, we aim to extend CTPO to broader agentic settings involving longer-horizon interaction and richer environment feedback.

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A Proofs for Section 3

A.1 Proof for Proposition 1

Proof. It suffices to show that each term in the sum is unbiased. For position t , we write A_t as shorthand for $A_t(s_t, a_t)$. Since the transition dynamics are deterministic, $s_{t'}$ is fully determined by the prompt x and the preceding actions $a_{1:t'-1}$ for all t' . Therefore the prefix likelihood factorizes as $\pi_\theta(a_{1:t} | x) = \prod_{t'=1}^t \pi_\theta(a_{t'} | s_{t'})$, and similarly for π_b . Applying the change of measure to the prefix:

$$\begin{aligned}
& \mathbb{E}_{\tau \sim \pi_b} [\rho_t^{\text{cum}} A_t \nabla_\theta \log \pi_\theta(a_t | s_t)] \\
&= \mathbb{E}_{\tau \sim \pi_b} \left[\frac{\pi_\theta(a_{1:t} | x)}{\pi_b(a_{1:t} | x)} A_t \nabla_\theta \log \pi_\theta(a_t | s_t) \right] \\
&= \mathbb{E}_{a_{1:t} \sim \pi_\theta(\cdot | x)} [A_t \nabla_\theta \log \pi_\theta(a_t | s_t)],
\end{aligned}$$

where the last step follows from the change of measure identity, and we have marginalized over the suffix $a_{t+1:H}$, which does not affect A_t or $\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ as both depend only on $(x, a_{1:t})$. This equals $\mathbb{E}_{\tau \sim \pi_{\theta}}[A_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$, which is the correct policy gradient term at position t . $\square$

A.2 Proof for Proposition 2

Proof. (i) Decompose the sequence-level ratio as $\rho^{\text{seq}} = \rho_t^{\text{cum}} \cdot \epsilon_t$, where $\epsilon_t = \prod_{t'=t+1}^H r_{t'}$ is the suffix ratio. By the likelihood ratio identity, $\mathbb{E}_{\pi_b}[\epsilon_t | a_{1:t}] = 1$. Applying the law of total variance:

$$\begin{aligned} \text{Var}(\rho^{\text{seq}}) &= \text{Var}(\mathbb{E}[\rho_t^{\text{cum}} \epsilon_t | a_{1:t}]) + \mathbb{E}[\text{Var}(\rho_t^{\text{cum}} \epsilon_t | a_{1:t})] \\ &= \text{Var}(\rho_t^{\text{cum}}) + \mathbb{E}[(\rho_t^{\text{cum}})^2 \text{Var}(\epsilon_t | a_{1:t})], \end{aligned}$$

where the second equality uses the fact that ρ_t^{cum} is measurable with respect to $a_{1:t}$ and $\mathbb{E}[\epsilon_t | a_{1:t}] = 1$. If π_{θ} and π_b differ at some future state reachable from $a_{1:t}$ with positive probability under π_b , then $\text{Var}(\epsilon_t | a_{1:t}) > 0$. Therefore, if this occurs on a set of prefixes with positive probability, the second term is strictly positive, yielding $\text{Var}(\rho^{\text{seq}}) > \text{Var}(\rho_t^{\text{cum}})$.

(ii) Under the independence assumption, the second moment factorizes as:

$$\mathbb{E}_{\pi_b}[(\rho_t^{\text{cum}})^2] = \prod_{t'=1}^t \mathbb{E}_{\pi_b}[r_{t'}^2].$$

For each position t' , we have

$$\begin{aligned} \mathbb{E}_{\pi_b}[r_{t'}^2] &= \mathbb{E}_{\pi_b} \left[\sum_{a \in \mathcal{V}} \pi_b(a | s_{t'}) \left(\frac{\pi_{\theta}(a | s_{t'})}{\pi_b(a | s_{t'})} \right)^2 \right] \\ &= 1 + \mathbb{E}_{\pi_b} [\chi^2(\pi_{\theta}(\cdot | s_{t'}) \| \pi_b(\cdot | s_{t'}))] = 1 + \chi_{t'}^2. \end{aligned}$$

Thus,

$$\mathbb{E}_{\pi_b}[(\rho_t^{\text{cum}})^2] = \prod_{t'=1}^t (1 + \chi_{t'}^2).$$

Since $\mathbb{E}_{\pi_b}[\rho_t^{\text{cum}}] = 1$ by the likelihood-ratio identity, subtracting 1 yields

$$\text{Var}(\rho_t^{\text{cum}}) = \prod_{t'=1}^t (1 + \chi_{t'}^2) - 1.$$

Applying the same argument to ρ^{seq} yields the stated expression for $\text{Var}(\rho^{\text{seq}})$. $\square$